Iowa State University

# LAB 8 - BUCKLING/CRIPPLING REPORT

Lucas Tavares Vasconcellos

Professor

Dr. Vinay Dayal

*College of Engineering*

*Aerospace Engineering*

*Aer E 4260 Design of Aerospace Structures*

Ames, November 2024

# Contents

# 1

# Introduction

This report examines the buckling behavior and crippling load capacity of aluminum structural elements under compressive loads. The analysis involves two problems. In Problem 1, we analyze a section made of 7075-T6 aluminum to determine its crippling stress and maximum compressive axial load capacity. In Problem 2, we estimate the crippling load of a combined Z-beam and skin structure in a wing.

# 2

# Figures and Tables

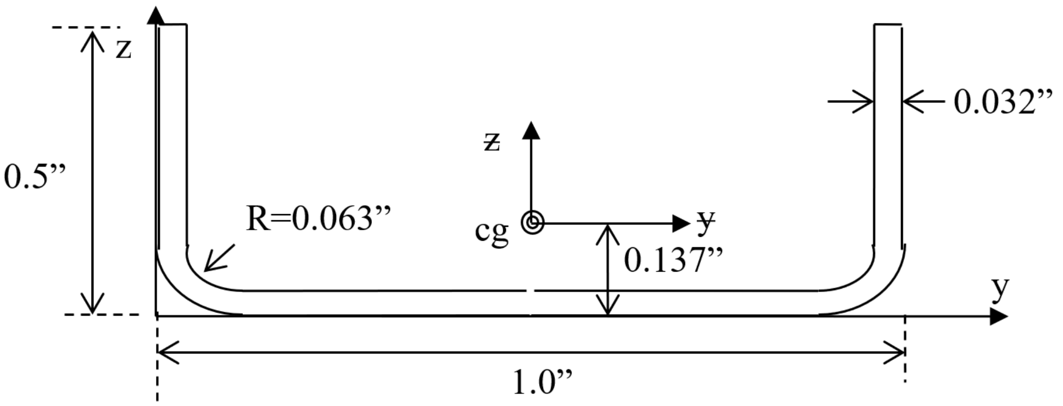


**Figure 2.1:** *Problem 1 Section*

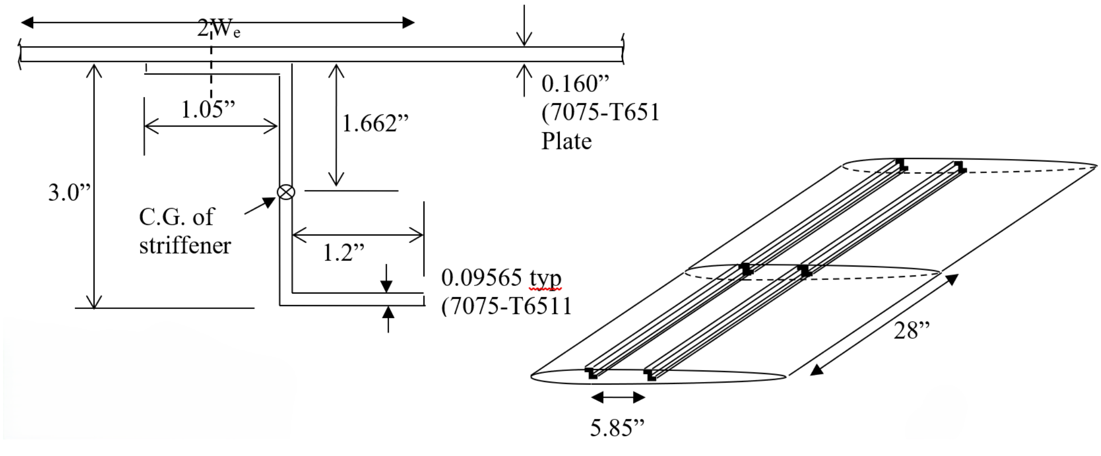


**Figure 2.2:** *Problem 2: Stiffener and skin of a wing; Cross-section and isometric views*

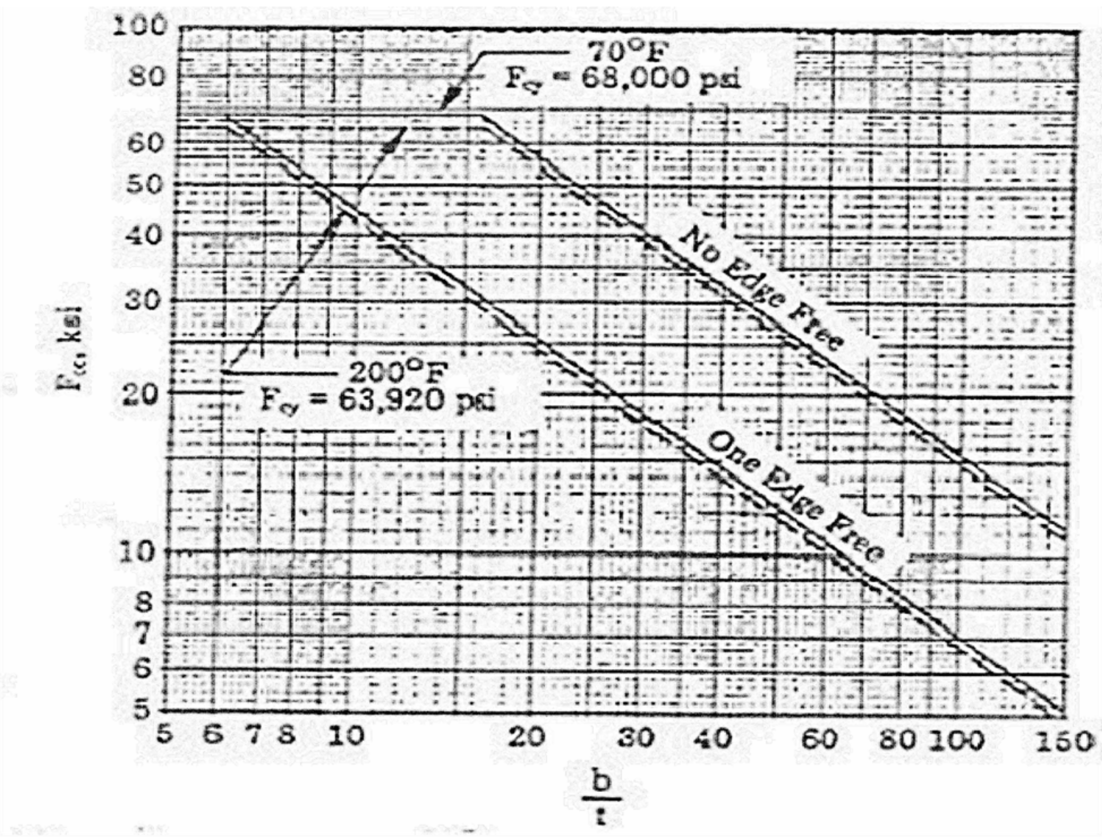


**Table 2.1:** *Compressive crippling of formed sections, 7075-T6,-T651, Bare aluminum*

# 3

# Methodology and Results

## 3.1 Problem 1

We start problem 1 by creating a table with $t$, $b$, $b/t$, and $F_{cc}$ values for each segment of the section (1 is the left segment with free end, 2 is the right segment with free end, and 3 is the segment with no free end). With $b/t$, the $F_{cc}$ value was retrived from Table 2.1 for each segment.

| Segment | $t$ (in) | $b$ (in) | $b/t$ | $F_{cc}$ (ksi) |
|---|---|---|---|---|
| 1 | 0.032 | 0.5 | 15.625 | 31 |
| 2 | 0.032 | 0.5 | 15.625 | 31 |
| 3 | 0.032 | 0.936 | 29.25 | 42 |

With the known values of $F_{cc}$ for each segment, we could calculate $F_{cc}$ for the entire section:

$$F_{cc} = \frac{2(0.5\text{ in})(0.032\text{ in})(31\text{ ksi}) + (0.936\text{ in})(0.032\text{ in})(42\text{ ksi})}{2(0.5\text{ in})(0.032\text{ in}) + (0.936\text{ in})(0.032\text{ in})} = 36.318\text{ ksi} \tag{3.1}$$

We also need I:

$$A_{1,2} = (0.032\text{ in})(0.5\text{ in}) = 0.016\text{ in}^2 \quad , \quad A_3 = (0.032\text{ in})(0.936\text{ in}) = 0.030\text{ in}^2 \tag{3.2}$$

$$I_{yy,1,2} = \frac{b \cdot h^3}{12} + A_{1,2} \cdot dz^2 = \frac{(0.032\text{ in})(0.5\text{ in})^3}{12} + (0.016\text{ in}^2)\left(\frac{0.5\text{ in}}{2} - 0.137\text{ in}\right)^2 = 0.000538\text{ in}^4 \tag{3.3}$$

$$I_{yy,3} = \frac{(0.936\text{ in})(0.032\text{ in})^3}{12} + (0.030\text{ in}^2)\left(0.137\text{ in} - \frac{0.032\text{ in}}{2}\right)^2 = 0.000518\text{ in}^4 \tag{3.4}$$

$$I_{yy} = 2I_{yy,1,2} + I_{yy,3} = 2(0.000538\text{ in}^4) + (0.000518\text{ in}^4) = 1.594 \times 10^{-3}\text{ in}^4 \tag{3.5}$$

$$I_{zz,1,2} = \frac{h \cdot b^3}{12} + A_{1,2} \cdot dy^2 = \frac{(0.5\text{ in})(0.032\text{ in})^3}{12} + (0.016\text{ in}^2)\left(\frac{1\text{ in} - 0.032\text{ in}}{2}\right)^2 = 0.003791\text{ in}^4 \tag{3.6}$$

$$I_{zz,3} = \frac{(0.032\text{ in})(0.936\text{ in})^3}{12} = 0.002187\text{ in}^4 \tag{3.7}$$

$$I_{zz} = 2I_{zz,1,2} + I_{z,3} = 2(0.003791\text{ in}^4) + (0.002187\text{ in}^4) = 9.768 \times 10^{-3}\text{ in}^4 \tag{3.8}$$

In between $I_{yy}$ and $I_{zz}$, we chose $I_{yy}$, since it is smaller (limiting the most). Now, we are able to calculate $\rho$ and, consequently, $F_{cr}$:

$$A_{total} = 2(0.016\text{ in}^2) + (0.030\text{ in}^2) = 0.062\text{ in}^2 \tag{3.9}$$

$$\rho = \sqrt{\frac{1.594 \times 10^{-3}\text{ in}^4}{0.062\text{ in}^2}} = 0.160\text{ in} \tag{3.10}$$

$$F_{cr} = (36.318\text{ ksi})\left(1 - \frac{(36.318\text{ ksi})}{4 \cdot \pi^2 \cdot (10.7 \times 10^3\text{ ksi})}\left(\frac{10\text{ in}}{0.16\text{ in}}\right)^2\right) = 24.121\text{ ksi} \tag{3.11}$$

And the maximum compressive axial force $f_{c,max}$ is:

$$f_{c,max} = F_{cr} \cdot A_{total} = (24.121\text{ ksi})(0.062\text{ in}^2) = 1495\text{ lbs} \tag{3.12}$$

## 3.2 Problem 2

For problem 2, we need to estimate the crippling load of a combination of a Z-beam connected to a skin in a wing, as seen in Figure 2.2. A MATLAB code (Appendix B) was developed to iterate over the following steps (similar procedure to Problem 1) and report final values when $W_e$ converges. The initial valued assume for $W_e$ is 3 in.

1. Calculate $I$ for stiffener + skin
2. Calculate $F_{cc}$ for stiffener + skin
3. Calculate $F_{cr}$
4. Calculate $W_{e,new}$ from $F_{cr}$
5. Update $W_e$ and go back to Step 1
6. Iterate until error < 0.01

The final reported values were:

| Iteration | $W_e$ | $I$ | $F_{cc}$ | $A$ | $F_{cr}$ |
|---|---|---|---|---|---|
| 0 | 3 in | 3.3219 in$^4$ | 36.6976 ksi | 1.4622 in$^2$ | 35.5974 ksi |
| 1 | 2.3579 in | 2.7539 in$^4$ | 40.5776 ksi | 1.2567 in$^2$ | 39.1831 ksi |
| 2 | 2.2474 in | 2.6562 in$^4$ | 42.3319 ksi | 1.2213 in$^2$ | 40.8026 ksi |
| 3 | 2.2024 in | 2.6163 in$^4$ | 42.3119 ksi | 1.2069 in$^2$ | 40.7791 ksi |

# 4

# Conclusion

The analysis successfully determined the crippling loads and buckling characteristics of the aluminum structures in both problems. For Problem 1, the calculated crippling stress value was 24.121 ksi and the maximum compressive axial force value was 1495 lbs. In Problem 2, the iterative process resulted in a converged $W_e$ = 2.2024 in, with a $F_{\mathrm{cr}}$ = 40.7791 ksi for the Z-beam and skin combination.

# A

# Appendix A

## A.1 Rough Work

AerE 426

Lab 8 Assignment

| (1)(A) | | t (in) | b (in) | b/t | Fcc (ksi) |
|---|---|---|---|---|---|
| one edge free | 1 | 0.032 | 0.5 | 15.625 | 31 |
| one edge free | 2 | 0.032 | 0.5 | 15.625 | 31 |
| no edge free | 3 | 0.032 | 0.936 | 29.25 | 42 |

$$F_{cc,total} = \frac{2(0.5\,in)(0.032\,in)(31\,ksi) + (0.936\,in)(0.032\,in)(42\,ksi)}{2(0.5\,in)(0.032\,in) + (0.936\,in)(0.032\,in)}$$

$F_{cc,total} = 36.318\,ksi$ //

$$I_{yy1} = \frac{bh^3}{12} + A_1 d_{z1}^2 = \frac{(0.032\,in)(0.5\,in)^3}{12} + (0.032\,in)(0.5\,in)\left(\frac{0.5\,in}{2} - 0.137\,in\right)^2$$

$I_{yy1} = I_{yy3} = 0.000538\,in^4$ //

$$I_{yy2} = \frac{(0.936\,in)(0.032\,in)^3}{12} + (0.032\,in)(0.936\,in)\left(0.137\,in - \frac{0.032\,in}{2}\right)^2$$

$I_{yy2} = 0.000518\,in^4$

(smaller) $I_{yy} = \sum_i I_{yyi} = 1.594 \times 10^{-3}\,in^4$ //

$A_{1,3} = (0.5\,in)(0.032\,in)$

$A_{1,3} = 0.016\,in^2$

$A_2 = (0.936\,in)(0.032\,in)$

$A_2 = 0.0300\,in^2$

$$I_{zz1} = I_{zz3} = \frac{h_{1,3} b_{1,3}^3}{12} + A_{1,3} \cdot d_{y1,3}$$

$$= \frac{(0.5\,in)(0.032\,in)^3}{12} + (0.016\,in^2)\left(\frac{1\,in}{2} - 0.032\,in\right)^2$$

$= 0.003791\,in^4$

$$I_{zz2} = \frac{(0.032\,in)(0.936\,in)^3}{12} = 0.002187\,in^4$$

$I_{zz} = \sum_i I_{zzi} = 9.768 \times 10^{-3}\,in^4$ //

$A_t = \sum_i A_i = 0.062\,in^2$ //

$$\rho = \sqrt{\frac{I_{yy}}{A_t}} = \sqrt{\frac{(1.594 \times 10^{-3}\,in^4)}{(0.062\,in^2)}} = 0.160\,in$$

$F_{cr} = (36.318\,ksi)\left(1 - \frac{(36.318\,ksi)}{4\pi^2(10.7\times10^3\,ksi)}\left(\frac{10\,in}{0.16\,in}\right)^2\right)$

$\boxed{F_{cr} = 24.121\ ksi}$

(B). $F_{c,max} = F_{cr}\cdot A_t = (24.121\,ksi)(0.062\,in^2)$

$\boxed{F_{c,max} = 1495\ lbs}$

2 (1) Assume $W_e = 3\,in$

(2). $I_{stiff} = 0.6681\ in^4$

$I_{skin} = \frac{(2W_e)(0.160\,in)^3}{12} + (2W_e)(0.160\,in)(1.662\,in)^2$

$I_{skin} = 2.6538\,in^4$ $\quad I = I_{stiff} + I_{skin} = 3.2835\,in^4$

| (3). | t (in) | b (in) | b/t | $F_{cc}$ (ksi) |
|---|---|---|---|---|
| 1 | 0.160 | 2We | 2We/0.160 | |
| 2 | 0.09565 | 1.05 | 11.0 | 42 |
| 3 | 0.09565 | 3 | 31.4 | 40 |
| 4 | 0.09565 | 1.2 | 12.5 | 38 |

(5) After 2 iterations, We converged to 2.9 in

it 1: $F_{cc,skin} = 16\ ksi$ ; it 2: $F_{cc,skin} = 16\ ksi$

B

# Appendix B

## B.1 MATLAB Code

```
% AerE 426, Lab Assignment 8 - Buckling: Problem 2, Lucas Tavares

clear, clc, close all

% Problem variables
L = 28; %in
E = 10.7e3; %ksi

% Assume initial value for We
We = 3; %in

% Calculate I_stiffner
I_stiffner = (1.05 * (0.09565^3) / 12 + (1.05 * 0.09565) * (1.662 - 0.09565
    / 2)^2) + (1.2 * (0.09565^3) / 12 + (1.2 * 0.09565) * (3 - 1.662 -
    0.09565 / 2)^2) + (0.09565 * 3^3 / 12); %in^4

% Loop variables
error = 1;
it = 0;

while it < 4

    fprintf('\nIteration %d: \n', it+1)

    % Calculate I_skin and total I
    I_skin = (2 * We) * 0.160^3 / 12 + (2 * We) * 0.160 * 1.662^2; %in^4
    I = I_stiffner + I_skin; %in^4

    % Calculate rho
    A = (2 * We) * 0.160 + 1.05 * 0.09565 + 3 * 0.09565 + 1.2 * 0.09565; %in
        ^2
    rho = sqrt(I / A);

    % calculate b/t and input Fcc for the skin
    b_over_t_skin = 2 * We / 0.160;
```

```
    fprintf('b/t for the skin = %.01f \n', b_over_t_skin)
    Fcc_skin = input('Please enter Fcc for the skin: ');

    % Calculate Fcc
    sum_b_t = (2 * We) * 0.160 + 1.05 * 0.09565 + 3 * 0.09565 + 1.2 *
        0.09565; %in^2
    sum_b_t_Fcc = (2 * We) * 0.160 * Fcc_skin + 1.05 * 0.09565 * 42 + 3 *
        0.09565 * 40 + 1.2 * 0.09565 * 38; %kip
    Fcc = sum_b_t_Fcc / sum_b_t; %ksi

    % Calculate Fcr
    Fcr = Fcc * (1 - (Fcc / (4 * pi^2 * E)) * (L / rho)^2); %ksi

    % Calculate We
    We_new = 0.85 * 0.160 * sqrt(E / Fcr); %in

    % Calculate error
    error = abs(We - We_new)/We;
    We = We_new; %in
    it = it + 1;

    % Print iteration results
    fprintf('I = %.04f in^4 \n', I)
    fprintf('Fcc = %.04f ksi \n', Fcc)
    fprintf('A = %.04f in^2 \n', A)
    fprintf('Fcr = %.04f ksi \n', Fcr)
    fprintf('Error = %.04f\n', error)
    fprintf('New We = %.04f in \n', We)

end
```